MORSE HOMOLOGY AND INSTANTON FLOER HOMOLOGY

LILY ZHANG

ABSTRACT. I gave two short talks in the graduate topics course in geometry, *Gauge Theory*. These are the notes from those talks.

1. Morse Homology

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Let X be a smooth, closed manifold. (H_*^{Morse} denotes the homology of the Morse chain complex for a Morse–Smale pair (f, g) on X, built from critical points of f and the negative gradient flow of f w.r.t. g.)

We want to show that there is a canonical isomorphism $H_*^{\text{Morse}}(f, g) \simeq H_*(X)$.

Define a chain map $D\colon C_*^{\text{Morse}}(f, g) \to C_*(X)$ by sending critical points to their descending manifolds. Define $A\colon C_*(X) \to C_*^{\text{Morse}}(f, g)$.

We want to show:

(1) $A \cdot D = \mathrm{id}$ on the chain level.
(2) $D \cdot A$ is chain homotopic to the identity.

$\Rightarrow$ D_* is an isomorphism on homology.

Definition 1.1. First, we define $C_*(X)$. An i-simplex σ is generic if it is smooth and each face of σ is transverse to the ascending manifolds of all critical points.

$C_i(X)$ is the subgroup of i-dimensional currents generated by such generic simplices.

Construction 1.2. For (1), we define the compactification of the descending manifold $D(p)$.

$D(p) \xrightarrow{\text{compactify}} \overline{D(p)}$: a manifold with corners.

The codimension k stratum is:

$$\overline{D(p)}_k = \bigcup_{\substack{q_1,\ldots,q_k \in \mathrm{Crit}(f) \\ p,q_1,\ldots,q_k \text{ distinct}}} M(p, q_1) \times M(q_1, q_2) \times \cdots \times M(q_{k-1}, q_k) \times D(q_k).$$

The boundary (the codimension-1 stratum) is:

$$\partial\overline{D(p)} = \bigcup_{p \neq q} (\pm)\, M(p, q) \times D(q).$$

The projection to the last factor $D(q_k) \subset X$ glues to a smooth map $e\colon \overline{D(p)} \to X$ (extending the inclusion $D(p) \hookrightarrow X$). A point of $\overline{D(p)}$ is either a point $x \in D(p)$ or a broken object. For a broken object, the first flow line goes from p to q, and then the remaining point lies in $D(q)$.

Because $\overline{D(p)}$ is a compact, oriented manifold with corners, it has a fundamental current $[\overline{D(p)}]$. Define $D(p) := e_*\big([\overline{D(p)}]\big) \in C_{\mathrm{ind}(p)}(X)$. Extend linearly to get $D\colon CM_*^{\text{Morse}}(f, g) \to C_*(X)$.

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Lemma 1.3 (Hutchings, Lemma 3.5)**.** *D is a chain map, i.e.* $\partial D = D\,\partial_{\text{Morse}}$.

Proof sketch. The boundary consists of broken trajectories.

$$\partial\overline{D(p)} = \bigcup_{p\neq q} (\pm)\, M(p,q)\times D(q).$$

Pushing forward by e turns this into a sum of currents:

$$\partial D(p) = \sum_{p\neq q} (\pm)\, e_*\big[\,\overline{M(p,q)\times D(q)}\,\big] \in C_{i-1}(X), \qquad p\in \mathrm{Crit}_i(f), \quad (\pm) = (-1)^{\,i+\mathrm{ind}(q)+1}.$$

(1) $\mathrm{ind}(q) > i-1$: $M(p,q)$ is empty for dimension reasons.
(2) $\mathrm{ind}(q) < i-1$: $M(p,q)\times D(q)$ maps under e into the support of $D(q)$.
The support has dimension $i-2$, so, as an $i-1$ current, it contributes zero.

$\Rightarrow$ q can contribute only when $\mathrm{ind}(q) = i-1$.

If $\mathrm{ind}(q) = i-1$ ($q\in \mathrm{Crit}_{i-1}(f)$), $M(p,q)$ is 0-dimensional.
Thus

$$e_*[M(p,q)\times D(q)] = \#M(p,q)\cdot e_*[\overline{D(q)}] = \#M(p,q)\cdot D(q).$$

$$\begin{aligned}\partial D(p) &= \sum_{q\in \mathrm{Crit}_{i-1}(f)} \#M(p,q)\cdot D(q)\\ &= D\Big(\sum_{q\in \mathrm{Crit}_{i-1}(f)} \#M(p,q)\cdot q\Big) \qquad \rightsquigarrow D \text{ is a chain map.}\\ &= D(\partial_{\text{Morse}}\, p).\end{aligned}$$

□

Definition 1.4. Define the chain map A. Let σ be a generic i-simplex, and let q be a critical point.

$M(\sigma,q)$ is the moduli space of gradient flow lines from σ to q; i.e. of maps $\gamma\colon [0,\infty)\to X$ such that $\gamma(0)\in\sigma$, $\gamma'(s) = -\nabla f(\gamma(s))$, and $\lim_{s\to\infty}\gamma(s) = q$.

Orient $M(\sigma,q)$ via the isomorphism $T_{\gamma(0)}\sigma \cong T_\gamma M(\sigma;q)\oplus T_q D(q)$, declaring it orientation-preserving. Since $\dim M(\sigma;p) = i-\mathrm{ind}(p)$, it is 0-dimensional if $\mathrm{ind}(p) = i$. Then define

$$A(\sigma) := \sum_{p\in \mathrm{Crit}_i(f)} \#M(\sigma,p)\,p.$$

Extend linearly to get $A\colon C_i(X)\to CM_i^{\text{Morse}}(f,g)$.

$$\begin{cases} A \text{ is a chain map}: \ A\partial = \partial^{\text{Morse}}A.\\ A\cdot D = \mathrm{id}\colon C_i^{\text{Morse}} \longrightarrow C_i^{\text{Morse}}.\\ D\cdot A \text{ is chain homotopic to id on } C_*(X)\end{cases}$$
$$\Rightarrow \quad H_*^{\text{Morse}}(f,g)\cong H_*(C_*(X))\cong H_*(X).$$

2. Instanton Floer Homology

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$$\underbrace{I_*(Y)}_{(1)} \rightsquigarrow \underbrace{I_\bullet(Y)}_{(2)} \longrightarrow \ell(Y) \rightsquigarrow \text{cobordism inequality}$$

(1) ordinary instanton homology is a 3-manifold invariant.
(2) filtered IP-module $\to$ defines $\ell(Y)$; cobordism maps shift this invariant.

Proposition 2.1.

(1) If Y is an integer homology sphere, $H_*(Y;\mathbb{Z}) \cong H_*(S^3;\mathbb{Z})$, then there is an associated $\mathbb{Z}/8$-graded module $I_d(Y;R)$, $d \in \mathbb{Z}/8$. $I_d(Y;R)$ is an orientation-preserving diffeomorphism invariant and is functorial under cobordisms $(W,c)\colon Y \to Y'$ with $b_1(W) = b^+(W) = 0$, where c is a closed, oriented, embedded surface.

(2) We also have the vanishing $I_d(S^3;R) = 0$.

(3) There is an admissible version for manifolds with the homology of $S^2 \times S^1$: $I_d^w(Y;R)$ is graded by $\mathbb{Z}/4$, and $I_d^w(S^2 \times S^1;R) = 0$.

Next, we briefly discuss how $I_*(Y)$ enriches to an IP-module $I_\bullet(Y)$ if Y is an integer homology sphere.

Remark 2.2. $I_*(Y)$ is $\mathbb{Z}/8$-graded, so it consists of groups $I_d(Y)$, $d \in \mathbb{Z}/8$. But $I_\bullet(Y)$ contains groups $F_r I_d(Y)$ for real filtration levels r and gradings $d \in \mathbb{Z}$. If $r = \infty$, $F_\infty I_d(Y) \cong I_{d \bmod 8}(Y)$.

Also, an IP-module has periodicity maps $\phi_{r,d}\colon F_r I_d(Y) \to F_{r+1} I_{d+8}(Y)$. These say that shifting the integer grading by 8 shifts the filtration by 1.

Definition 2.3. We define the numerical invariants of the IP-module associated with Y. Set $A_\bullet = I_\bullet(Y;\mathbb{F}_2)$. Then

$$\kappa_Y(d) = \inf\left\{ r \in \mathbb{R} \;\middle|\; i_r^\infty\colon F_r I_d(Y;\mathbb{F}_2) \longrightarrow F_\infty I_d(Y;\mathbb{F}_2) \text{ is nonzero} \right\}.$$

$$\ell(Y) = \ell\big(I_\bullet(Y;\mathbb{F}_2)\big) = \inf_{d\in\mathbb{Z}} \underbrace{\left(\kappa_Y(d) - \frac{d}{8}\right)}_{\text{corrects the IP periodicity}} .$$

$\ell(Y)$ is also an orientation-preserving diffeomorphism invariant; this follows from $I_\bullet(Y)$. $\ell(Y) < \infty$ when ordinary instanton homology is nonzero in some mod-8 grading.

The next proposition says that cobordism maps are compatible with the IP-module structure.

Proposition 2.4. *In the same setup, there is an induced IP-morphism $(W,c)_*\colon I_\bullet(Y) \longrightarrow I_\bullet(Y')$. This means $(W,c)_*\colon F_r I_d(Y) \longrightarrow F_{r+L} I_{d+D}(Y')$ for all r and d. It is compatible with connecting maps i and periodicity maps ϕ.*

The degree is $D = -2c^2$, and the level is $L = -\frac{c^2}{4} - \eta(W,c)$. Here $\eta(W,c)$ is nonnegative. If $\pi_1(W) = 0$, then $\eta(W,c) > 0$.

If the induced map on ordinary instanton homology is injective (e.g. an isomorphism), then by the earlier lemma,

$$\ell(Y') \le \ell(Y) + L - \frac{D}{8}.$$

$$L - \frac{D}{8} = \left(\frac{-c^2}{4} - \eta(W,c)\right) - \frac{-2c^2}{8} \;\Rightarrow\; L - \frac{D}{8} = -\eta(W,c).$$

Under this injectivity assumption, if $\pi_1(W) = 0$, $\ell(Y') < \ell(Y)$.

Example 2.5. $Y = \Sigma(2,3,5)$, oriented as the boundary of the negative definite E_8 plumbing. $[\alpha], [\beta]$ are flat irreducible $SU(2)$-connections. Their CS values are $CS(\alpha) = \frac{1}{120}$, $CS(\beta) = \frac{49}{120}$. Their gradings are $i(\alpha) = 1$, $i(\beta) = 5$. Then

$$\kappa_{\Sigma(2,3,5)}(d) = \begin{cases} \dfrac{1}{120} + \dfrac{d-1}{8}, & d \equiv 1 \pmod 8 \\[2ex] \dfrac{49}{120} + \dfrac{d-5}{8}, & d \equiv 5 \pmod 8 \\[2ex] \infty, & \text{otherwise.} \end{cases}$$

$$\frac{1}{120} - \frac{1}{8} = -\frac{7}{60} \qquad \frac{49}{120} - \frac{5}{8} = -\frac{13}{60}, \qquad \text{so } \ell(\Sigma(2,3,5)) = -\frac{13}{60}.$$

References

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[2] M. Hutchings, *Lecture notes on Morse homology (with an eye towards Floer theory and pseudoholomorphic curves)*, unpublished lecture notes, University of California, Berkeley, 2002. Available at https://math.berkeley.edu/~hutching/teach/276-2010/mfp.ps.